

1.Display following CSS Box model using html and css

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>CSS Box Model</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <div class="container">
    <h1>CSS Box Model Visualization</h1>
    <div class="margin">
      <span class="label">Margin</span>
      <div class="border">
        <span class="label">Border</span>
        <div class="padding">
          <span class="label">Padding</span>
          <div class="content">
            <span>Content</span>
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```


2. Display items in multiple column using property like column-count, column-gap, column-rule to display the following.

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>CSS Multi-Column Layout</title>
  <link rel="stylesheet" href="multiple_column.css">
</head>
<body>
  <section class="wrap">
    <h2>Display items in multiple columns</h2>
    <div class="columns">
      <div class="item">Item 1 - Lorem ipsum dolor sit, amet consectetur adipisicing elit.
Asperiores saepe quo beatae expedita commodi error corporis harum impedit, eos aliquam, odit
cumque molestiae, repudiandae laboriosam. Deserunt ab aperiam enim totam!.</div>

      <div class="item">Item 2 - Lorem ipsum dolor sit amet consectetur adipisicing elit.
Distinctio illum incidunt placeat eaque, dicta magnam aperiam iure repellat itaque, asperiores
molestias, eligendi dolores. Voluptatem ab suscipit tenetur, nemo doloremque ducimus!.</div>

      <div class="item">Item 3 - Lorem ipsum dolor, sit amet consectetur adipisicing elit. Ullam
similique tempore iste amet quos cumque dignissimos deserunt libero vero. Enim, nihil!
Consequatur velit quaerat unde, doloremque placeat explicabo amet animi!.</div>

      <div class="item">Item 4 - Lorem ipsum dolor sit amet consectetur adipisicing elit. Dolores
tempore, deleniti iure, nihil magnam sit pariatur quasi suscipit consectetur nam qui eaque autem
illum praesentium repellendus animi nobis porro quas..</div>

      <div class="item">Item 5 - Lorem ipsum, dolor sit amet consectetur adipisicing elit.
Perspiciatis pariatur ut distinctio cumque veniam hic corrupti enim, asperiores itaque? Quo et
delectus quod repellendus fugiat aut praesentium ratione deserunt nobis!.</div>

      <div class="item">Item 6 - Lorem ipsum dolor sit amet consectetur adipisicing elit.
Nesciunt doloremque repellat reprehenderit dolore porro officiis, doloribus tempora, et accusamus,
explicabo dignissimos. Qui, facilis maxime officia neque repellat assumenda impedit illo..</div>
    </div>
  </section>
</body>
</html>
```

```

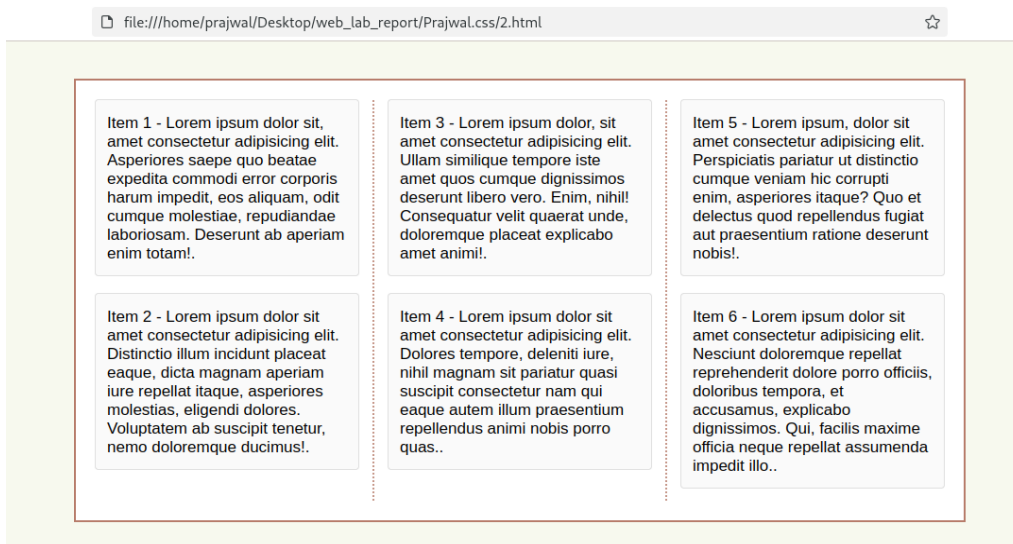
multiple_column.css
body {
  font-family: Arial, sans-serif;
  margin: 20px;
  background: #f2f7e9;
}

.container {
  border: 1px solid #d2691e;
  padding: 20px;
  column-count: 3;
  column-gap: 20px;
  column-rule: 1px dashed #aaa;
}

h2 {
  margin-bottom: 15px;
}

.item {
  border: 1px solid #ccc;
  padding: 10px;
  margin-bottom: 10px;
  background: #fff;
  break-inside: avoid;
}

```



. Use CSS selectors(universal selector, element-selector, element grouping selector, id and class selector) to display as follows.

HTML CODE.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Q3 – CSS Selectors Example</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <div id="main-content">
    <h1>Main Content Section</h1>

    <h2>Article Title</h2>
    <p>This paragraph is a direct child of the article (styled with child selector).</p>
    <div>
      <p>This paragraph is nested deeper (not styled with child selector).</p>
      <p class="highlight">This paragraph has the highlight class applied.</p>
    </div>

    <h3>Another Heading</h3>
    <p>This paragraph comes immediately after an h3 (styled with adjacent sibling selector).</p>
    <p>This paragraph doesn't come immediately after an h3.</p>

    <h4>Form Example</h4>
    <form>
      <label for="name">Name:</label>
      <input type="text" id="name" name="name"><br><br>
      <label for="email">Email:</label>
      <input type="email" id="email" name="email">
    </form>

    <p>Visit our <a href="#">website</a> for more information (hover to see effect).</p>
  </div>
</body>
</html>
```

3. style.css

```
body {  
    font-family: Arial, sans-serif;  
    margin: 40px;  
}
```

```
.container {  
    border: 1px solid #999;  
    padding: 20px;  
    background: #e6f2f9;  
    width: 70%;  
    margin: auto;  
}
```

```
h2 {  
    color: #004080;  
}
```

```
h3 {  
    color: #0066cc;  
}
```

```
article > p {  
    background: #cceeef;  
    padding: 5px;  
}
```

```
h3 + p {  
    background: #cceeef;  
    padding: 5px;  
}
```

```
.highlight {  
    background: #ffe699;  
    padding: 5px;  
}
```

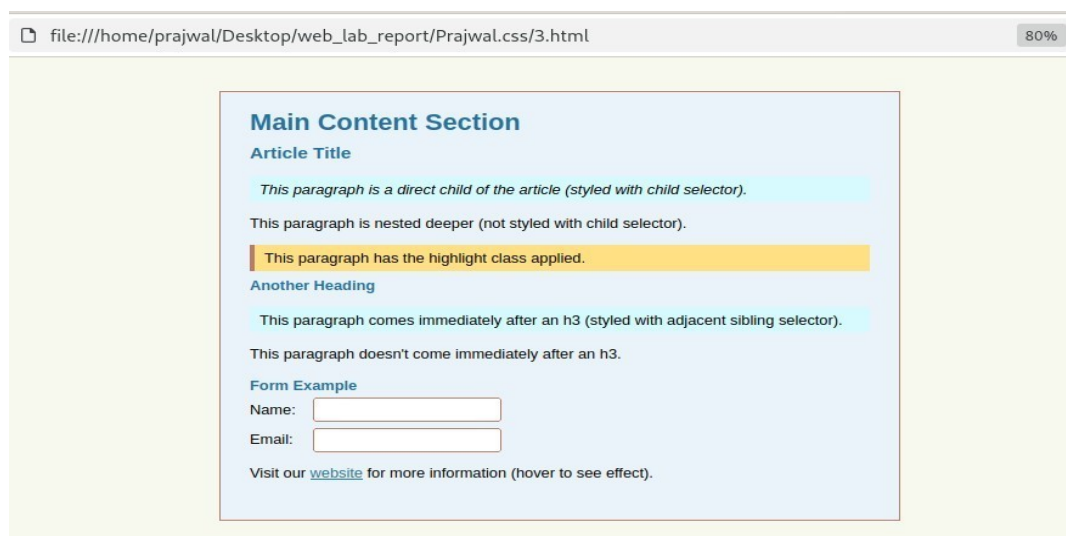
```
form {  
    margin-top: 15px;  
}
```

```
label {  
    display: inline-block;  
    width: 60px;  
    margin-bottom: 8px;  
}
```

```
input {  
    padding: 5px;  
    border: 1px solid #777;  
    margin-bottom: 10px;  
}
```

```
a {  
    color: #0040ff;  
    text-decoration: none;  
}
```

```
a:hover {  
    text-decoration: underline;  
    color: #ff6600;  
}
```



4. Use CSS attribute selectors for different elements.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Attribute Selectors Sample</title>
  <link rel="stylesheet" href="attribute-demo.css">
</head>
<body>
  <h3>CSS Attribute Selectors for Different Elements</h3>
  <div>
    <h4>1. [attribute] Selector</h4>
    <input type="text" placeholder="Required" data-required>
    <input type="text" placeholder="Optional">
    <p data-required>This paragraph has the data-required
attribute</p>
  </div>
  <div>
    <h4>2. [attribute="value"] Selector</h4>
    <button type="button">Regular Button</button>
    <button type="submit">Submit Button</button>
    <input type="submit" value="Submit Input">
  </div>
  <div>
    <h4>3. [attribute~="value"] Selector</h4>
    <p class="highlight">This paragraph has the highlight
class</p>
    <p>This paragraph doesn't have highlight</p>
  </div>
  <div>
    <h4>4. [attribute^="value"] Selector</h4>
    <p lang="en">English content</p>
    <p lang="en-US">American English content</p>
    <p lang="en-GB">British English content</p>
    <p lang="fr">French content (not selected)</p>
  </div>
</body>
</html>
```

```

4-attribute-demo.css
/* 1. [attribute] selector */
[data-required] {
border-left: 3px solid #e53e3e;
padding-left: 5px;
}

/* 2. [attribute="value"] selector */
[type="submit"] {
background: #26ec7c;
color: #fff;
border: none;
padding: 5px 12px;
border-radius: 3px;
}

/* 3. [attribute~="value"] selector */
.highlight {
background: #f1c40f;
padding: 2px 5px;
border-radius: 2px;
}

/* 4. [attribute^="value"] selector */
[lang^="en"] {
border-bottom: 2px dotted #9b59b6;
}

```

<
>
↺
🔖
📄 File /home/prajwal/Desktop/prajwal_css/4.html

CSS Attribute Selectors for Different Elements

1. [attribute] Selector

| This paragraph has the data-required attribute

2. [attribute="value"] Selector

3. [attribute~="value"] Selector

This paragraph has the highlight class

This paragraph doesn't have highlight

4. [attribute^="value"] Selector

English content

American English content

British English content

French content (not selected)

5. Use CSS pseudo elements selectors for different html elements.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Pseudo-element Selectors Demo</title>
  <link rel="stylesheet" href="pseudo-elements-demo.css">
</head>
<body>
  <h3>Use CSS pseudo-elements selectors for different HTML elements.</h3>

  <p class="before-after">This shows ::before and ::after.</p>

  <p class="first-letter">First letter is large and colored.</p>

  <p class="first-line">The first line of this paragraph will look different. Lorem ipsum dolor sit
amet, consectetur adipiscing elit. Etiam euismod, nunc at laoreet euismod, nulla.</p>

  <p class="selectable">Try selecting this text to see the ::selection effect!</p>
</body>
</html>
```

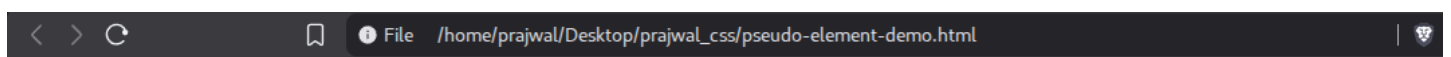
5-pseudo-element-demo.css

```
.before-after::before {
  content: "Before—";
  color: #1976d2;
  font-weight: bold;
}
.before-after::after {
  content: "—After";
  color: #e53935;
  font-style: italic;
}

.first-letter::first-letter {
  color: #388e3c;
  font-size: 2em;
  font-weight: bold;
}

.first-line::first-line {
  color: #8e44ad;
  background: #ecf0f1;
  font-weight: bold;
}

.selectable::selection {
  background: #ffe082;
  color: #d32f2f;
}
```



Use CSS pseudo-elements selectors for different HTML elements.

Before—This shows ::before and ::after.—After

First letter is large and colored.

The first line of this paragraph will look different. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam euismod, nunc at laoreet euismod, nulla.

Try selecting this text to see the ::selection effect!

.Use CSS pseudo-class selectors

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Anchor Pseudo-Classes Example</title>
```

```
  <link rel="stylesheet" href="pseudo-class-demo.css">
```

```
</head>
```

```
<body>
```

```
  <h3>6. Use CSS pseudo-class selectors for the following.</h3>
```

```
  <div class="pseudo-box">
```

```
    <div class="anchor-demo">
```

```
      <div class="anchor-title">Anchor Pseudo-Classes</div>
```

```
      <a href="https://claude.ai" target="_blank">Visit Claude.ai</a><br>
```

```
      <a href="https://maps.google.com" target="_blank">Visit Google Maps</a><br>
```

```
      <a href="https://google.com" target="_blank">Visit Google.com</a>
```

```
    </div>
```

```
    <div class="arrow-label">
```

```
      <span class="hover-arrow">→</span> &nbsp; <b>Hover</b>
```

```
      <span class="hover-note">Also add for unvisited, visited and active<br>for anchor pseudo
```

```
class</span>
```

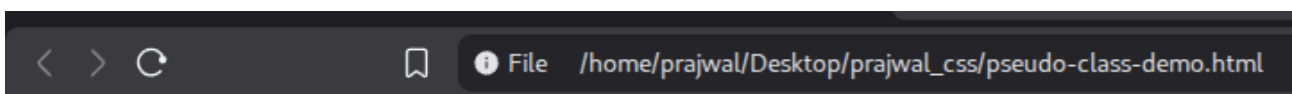
```
    </div>
```

```
  </div>
```

```
</body>
```

```
</html>
```

```
pseudo-class-demo.css
body {
  background: #eff8e5;
  font-family: Arial, sans-serif;
  padding: 0 18px;
}
ul {
  list-style: none;
  padding: 0;
}
a:link {
  color: #1c4dbd;
  text-decoration: underline;
}
a:visited {
  color: #b02e8d;
}
a:hover {
  color: #176b20;
  background: #ffffb3;
}
a:active {
  color: #d72121;
  background: #fdd;
}
small {
  color: #444;
  font-size: 0.9em;
}
b {
  font-size: 1.1em;
  color: #3e2170;
}
```



6. Use CSS pseudo-class selectors for the following.

Anchor Pseudo-Classes

[Visit Claude.ai](#)

[Visit Google Maps](#)

[Visit Google.com](#)

→ **Hover** Also add for unvisited, visited and active for anchor pseudo class

7. display property: HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Display Property Example</title>
  <link rel="stylesheet" href="display_property_demo.css">
</head>
<body>
  <div class="container">
    <h2>The display Property</h2>

    <div class="box">
      <h3>display: inline</h3>
      <p>
        Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat
        scelerisque elit sit amet consequat.
        Aliquam <span class="inline">Aliquam</span>
        <span class="inline">venenatis</span> gravida nisl sit amet facilisis.
        Nullam cursus fermentum velit sed laoreet.
      </p>

      <h3>display: inline-block</h3>
      <p>
        Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat
        scelerisque elit sit amet consequat.
        Aliquam <span class="inline-block">Aliquam</span>
        <span class="inline-block">venenatis</span> gravida nisl sit amet facilisis.
        Nullam cursus fermentum velit sed laoreet.
      </p>
    </div>

    <div class="box">
      <h3>display: block</h3>
      <p>
        Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat
        scelerisque elit sit amet consequat.
        <span class="block">Aliquam</span>
        <span class="block">venenatis</span> gravida nisl sit amet facilisis.
        Nullam cursus fermentum velit sed laoreet.
      </p>
    </div>
  </div>
</body>
</html>
```

7- display_property_demo.css

```
body {  
  font-family: Arial, sans-serif;  
  margin: 40px;  
}
```

```
.container {  
  display: flex;  
  justify-content: space-around;  
  border: 1px solid #d2691e;  
  padding: 20px;  
}
```

```
.sidebar {  
  width: 20%;  
  font-weight: bold;  
  font-size: 20px;  
}
```

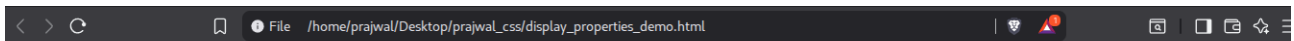
```
.content {  
  display: flex;  
  width: 75%;  
  border: 1px solid #8b0000;  
}
```

```
.box {  
  width: 50%;  
  border: 1px solid #8b0000;  
  padding: 15px;  
}
```

```
span.inline {  
  display: inline;  
  background: yellow;  
}
```

```
span.inline-block {  
  display: inline-block;  
  background: yellow;  
  padding: 5px;  
}
```

```
span.block {  
  display: block;  
  background: yellow;  
  padding: 5px;  
}
```



The display Property

display: inline

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat scelerisque elit sit amet consequat. Aliquam **Aliquam venenatis** gravida nisl sit amet facilisis. Nullam cursus fermentum velit sed laoreet.

display: inline-block

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat scelerisque elit sit amet consequat. Aliquam **Aliquam venenatis** gravida nisl sit amet facilisis. Nullam cursus fermentum velit sed laoreet.

display: block

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat scelerisque elit sit amet consequat.

Aliquam venenatis

gravida nisl sit amet facilisis. Nullam cursus fermentum velit sed laoreet.

. Try out the box-sizing: content-box | border-box; properties with different values and generalize its meaning.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Box Sizing and Box Decoration Break</title>
  <link rel="stylesheet" href="box_sizing.css">
</head>
<body>

  <div class="box content-box">
    <strong>box-sizing: content-box;</strong>
    <p><em>(width and height only apply to the content of the element)</em></p>
    <p>This div has a width of 400px, but the full width is 400px + 20px (left and right border)
      + 40px (left and right padding) = <strong>460px</strong>.</p>
  </div>

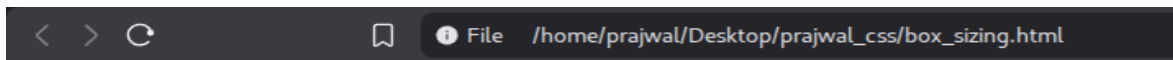
  <br>

  <div class="box border-box">
    <strong>box-sizing: border-box;</strong>
    <p><em>(width and height apply to all parts of the element: content, padding, and
borders)</em></p>
    <p>Here, the full width of the box is 400px, including content, padding, and borders.</p>
  </div>

</body>
</html>
```

box_sizing.css

```
body {  
    font-family: Arial, sans-serif;  
    margin: 40px;  
}  
  
.box {  
    width: 400px;  
    padding: 20px;  
    border: 10px solid green;  
    margin-bottom: 30px;  
    background-color: #f0f8ea;  
    text-align: center;  
    font-size: 18px;  
  
    line-height: 1.5;  
}  
  
.content-box {  
    box-sizing: content-box;  
    background-color: #ffeeee;  
}  
  
.border-box {  
    box-sizing: border-box;  
    background-color: #e6f7ff;}
```



box-sizing: content-box;

(width and height only apply to the content of the element)

This div has a width of 400px, but the full width is 400px + 20px (left and right border) + 40px (left and right padding) = **460px**.

box-sizing: border-box;

(width and height apply to all parts of the element: content, padding, and borders)

Here, the full width of the box is 400px, including content, padding, and borders.

Try out overflow, float and clear properties.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Overflow and Clear Examples</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <h2>Overflow and Clear Properties</h2>
  <div class="container">
    <div class="overflow-section">
      <h3>Overflow: visible</h3>
      <div class="box overflow-visible">
        <p>This content overflows the container and is still visible.</p>
      </div>
      <h3>Overflow: hidden</h3>
      <div class="box overflow-hidden">
        <p>This content is hidden outside the box. This content is hidden outside the box.</p>
      </div>
      <h3>Overflow: scroll</h3>
      <div class="box overflow-scroll">
        <p>This content is scrollable. This content is scrollable. This content is scrollable.</p>
      </div>
    </div>
    <!-- Right Side: Clear Examples -->
    <div class="clear-section">
      <h3>Clear: left</h3>
      <div class="float-box">
        <div class="float-left">Floated Left</div>
        <p class="clear-left">Cleared on the left.</p>
      </div>
      <h3>Clear: right</h3>
      <div class="float-box">
        <div class="float-right">Floated Right</div>
        <p class="clear-right">Cleared on the right.</p>
      </div>
      <h3>Clear: both</h3>
      <div class="float-box">
        <div class="float-left">Left</div>
        <div class="float-right">Right</div>
        <p class="clear-both">Cleared on both sides.</p>
      </div>
    </div>
  </div>
</body>
</html>
```

```

overflow_float.css
body {
    font-family: Arial, sans-serif;
    margin: 40px;
    text-align: center;
}

.container {
    display: flex;
    justify-content: space-between;
    align-items: flex-start;
    max-width: 1000px;
    margin: 0 auto;
    gap: 30px;
}

.overflow-section,
.clear-section {
    width: 45%;
}

.overflow-box {
    width: 200px;
    height: 70px;
    border: 1px solid #000;
    background: #f08080;
    margin: 20px auto;
    padding: 5px;
    text-align: left;
    font-size: 14px;
}

.visible {
    overflow: visible;
}

.hidden {
    overflow: hidden;
}

.scroll {
    overflow: scroll;
}

.clear-box {
    border: 2px dashed blue;
    background: #add8e6;
    margin: 20px 0;
    padding: 15px;
    text-align: left;
    overflow: hidden; /* to contain floats */
}

.float-left {
    float: left;
    width: 80px;
    height: 40px;
    background: lightgreen;
    margin: 5px;
    text-align: center;
    line-height: 40px;
}

.float-right {
    float: right;
    width: 80px;
    height: 40px;
    background: lightyellow;
}

margin: 5px;
text-align: center;
line-height: 40px;
}

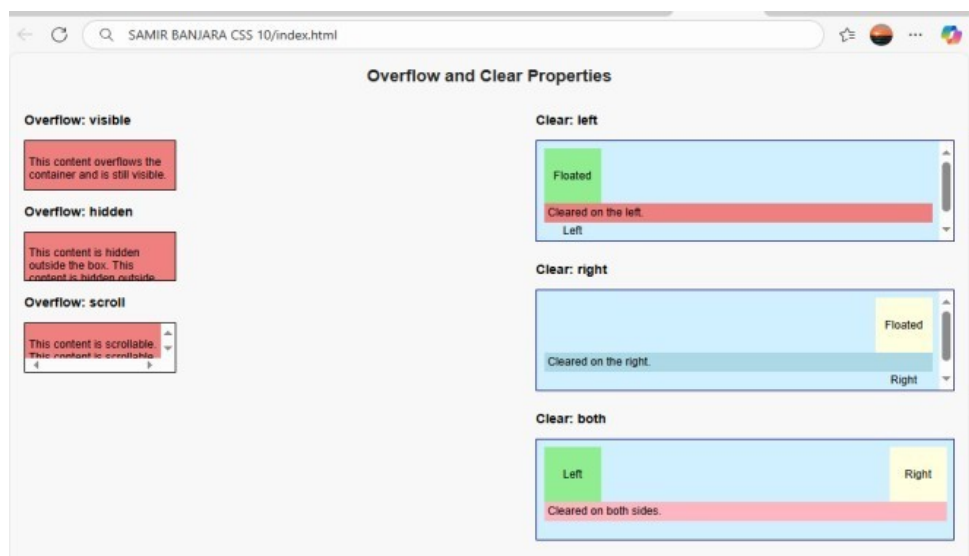
.clear-left {
    clear: left;
    background: lightcoral;
    padding: 10px;
    margin-top: 10px;
}

.clear-right {
    clear: right;
    background: lightcoral;
    padding: 10px;
    margin-top: 10px;
}

.clear-both {
    clear: both;
    background: lightcoral;
    padding: 10px;
    margin-top: 10px;
}

/* Optional: make scroll box
always show scrollbar for
demo */
.overflow-scroll {
    overflow: scroll;
}

```



```
flex_grid.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Responsive Layout — Flexbox</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="wrap">
  <div class="container">

    <header class="header card">
      <div class="label">Header</div>
    </header>

    <div class="row">
      <aside class="sidebar card">
        <div class="label">Sidebar</div>
      </aside>

      <main class="center">
        <nav class="nav card">
          <div class="label">Navigation</div>
        </nav>
        <section class="main card">
          <div class="label">Main</div>
        </section>
      </main>

      <aside class="rightbar card">
        <div class="label">ads</div>
      </aside>
    </div>

    <footer class="footer card">
      <div class="label">Footer</div>
    </footer>

  </div>
</div>

</body>
</html>
```

```

:root {
  --gap: 16px;
  --card-border: 2px solid #f6c2c2;
  --radius: 6px;
  --bg: #fff;
  --page-bg: #f7fbf6;
  --max-width: 1000px;
}

html,
body {
  height: 100%;
  margin: 0;
  background: var(--page-bg);
  font-family: Georgia, 'Times New Roman', serif;
  color: #222;
}

.wrap {
  max-width: var(--max-width);
  margin: 36px auto;
  padding: 18px;
}

.container {
  display: flex;
  flex-direction: column;
  gap: var(--gap);
}

.card {
  background: var(--bg);
  border: var(--card-border);
  border-radius: var(--radius);
  padding: 18px;
  box-sizing: border-box;
}

.header {
  height: 56px;
}

.row {
  display: flex;
  gap: var(--gap);
}

.sidebar {
  width: 160px;
  min-width: 120px;
}

.rightbar {
  width: 140px;
  min-width: 110px;
}

.center {
  flex: 1;
  display: flex;
  flex-direction: column;
  gap: var(--gap);
}

.nav {
  height: 48px;
}

.main {
  flex: 1;
}

.footer {
  height: 48px;
}

.card {
  border-color: #f4bebe;
}

.header .card,
.footer .card {
  display: flex;
  align-items: center;
}

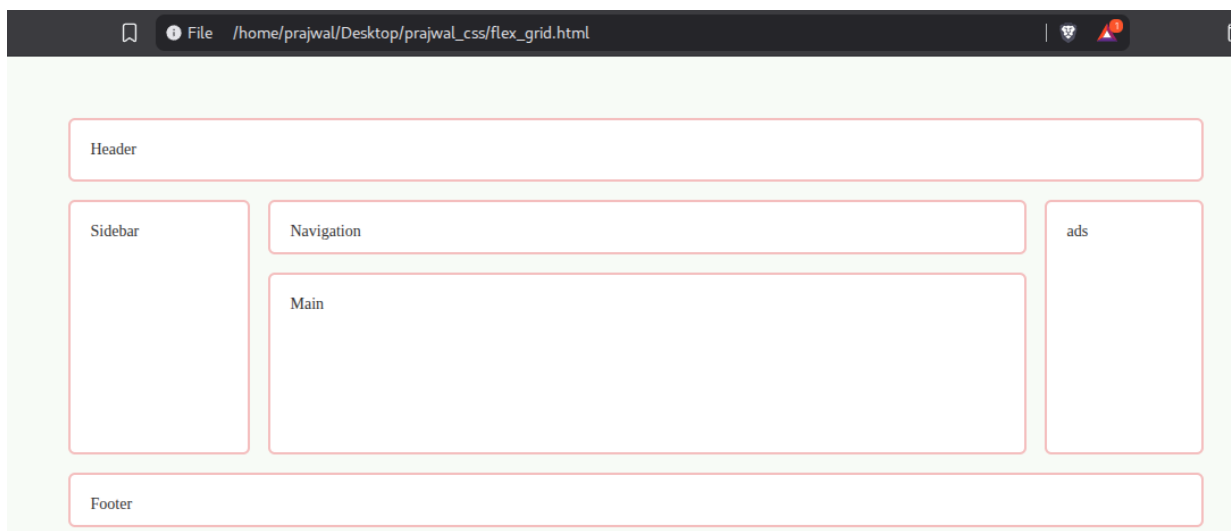
.label {
  font-size: 14px;
  color: #333;
}

@media (max-width: 800px) {
  .row {
    flex-direction: column;
  }

  .sidebar,
  .rightbar {
    width: auto;
  }

  .nav {
    order: 0;
  }
}

```



CSS- 18

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Home</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link href="https://cdn.jsdelivr.net/npm/font-awesome@4.7.0/css/font-awesome.min.css"
rel="stylesheet">
  <link rel="stylesheet" href="bootstrap.css">
</head>
<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#"><i class="fa fa-home"></i> MySite</a>
    <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#menu">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="menu">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link" href="#"><i class="fa fa-home"></i>
Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#"><i class="fa fa-phone"></i>
Contact</a></li>
      </ul>
    </div>
  </div>
</nav>

<section class="hero d-flex align-items-center justify-content-center text-center">
  <div>
    <h1>Welcome</h1>
    <p class="lead">Simple Responsive Homepage</p>
  </div>
</section>

<footer class="bg-dark text-white text-center p-3">
  <a href="#" class="text-white me-3"><i class="fab fa-linkedin"></i></a>
  <a href="#" class="text-white me-3"><i class="fab fa-facebook"></i></a>
  <a href="#" class="text-white"><i class="fab fa-instagram"></i></a>
  <p class="mb-0">&copy; 2025 MySite</p>
</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

CSS- bootstrap.css

```
.hero {  
  height: 70vh;  
  background: #007bff;  
  color: white;  
}
```

